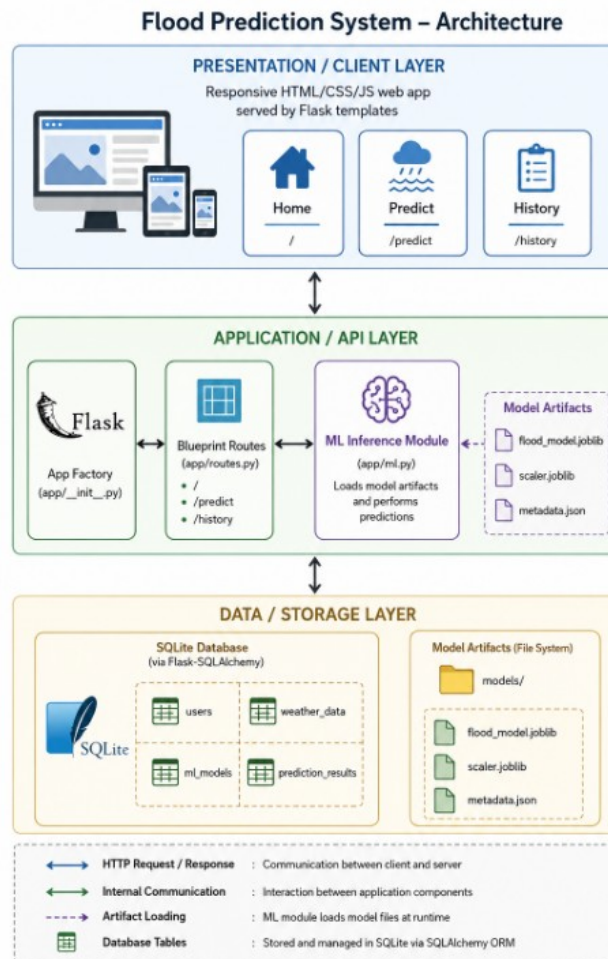


Solution Architecture

Date	15 July 2026
Team ID	XXXXXX
Project Name	FloodGuard - AI/ML Flood Prediction System
Maximum Marks	5 Marks

Solution Architecture Diagram



Component Description Table

Component Name	Description / Role	Technologies Used
Presentation Layer	Renders the home, prediction, and history pages; submits weather data via AJAX and displays results dynamically	HTML5, CSS3, JavaScript (Fetch API), Jinja2 templates
API / Routes Layer	Exposes /, /predict, /history and the POST /api/predict JSON endpoint; validates input and orchestrates the prediction flow	Flask 3.1, Flask Blueprints
ML Inference Module	Loads the trained model + scaler once at startup and runs predict_flood() for each request	scikit-learn / XGBoost, joblib, pandas
Training Pipeline	Offline pipeline that cleans data, trains 4 candidate models, compares them, and persists the best one	pandas, scikit-learn, XGBoost, matplotlib, seaborn

Database Layer	Persists users, weather submissions, registered models, and prediction results	SQLite, Flask-SQLAlchemy (ORM)
Deployment	Hosts the Flask app on a public URL, auto-deploying on every GitHub push	Render (Web Service), Gunicorn WSGI server, GitHub